

Local almost-everywhere constancy from the weak equation

MathlibAnnex.WeakDivergenceZero.nonempty_localAECOnstantAt

theorem

Turns constant mollifications into one almost-everywhere constant by choosing a common convergence point.

Statement

Let E be a finite-dimensional real normed vector space with its Borel structure and an additive Haar measure μ . Let $U \subseteq E$ be open and $u : E \rightarrow \mathbb{R}$ locally integrable on U . Assume

$$\int_U u(z) \operatorname{div} W(z) d\mu(z) = 0$$

for every C^1 vector field $W : E \rightarrow E$ for which a compact set K_W satisfies

$$\overline{\{z \in E : W(z) \neq 0\}} \subseteq K_W \subseteq U.$$

Here $\operatorname{div} W(z) := \operatorname{tr}(DW(z))$. For every $x \in U$, there are an open set V and a real number c with

$$x \in V \subseteq U, \quad u = c \quad \mu|_V\text{-almost everywhere.}$$

Assumptions

The domain need not be connected, bounded, or of finite measure. There is no global integrability, Lipschitz, or continuity assumption on u . No positive-dimension hypothesis is imposed. A point $x \in U$ is supplied, so the conclusion is local; if U is empty, there is no such point to consider.

Conclusion

The conclusion provides an open neighborhood V of the prescribed point x , with $V \subseteq U$, and a real constant c such that $u = c$ almost everywhere on V . It does not assert $u(x) = c$ at the originally supplied point, which might be exceptional.

The construction below uses the inner ball $B(x, R)$, the compact carrier $K = \overline{B}(x, 4R)$, and the intermediate ball $\overline{B}(x, 2R)$. All inclusions and compactness assertions follow from the chosen ball data and finite dimensionality. The same localized function and the same kernel sequence are used throughout.

Proof route

Localize to a compact ball. The weak equation makes every derivative of each smooth mollification zero on the inner ball, hence each mollification is constant there. Almost-everywhere convergence at one fixed good point forces these constants to converge. Uniqueness of limits then identifies the value at almost every other point.

▼ Proof steps

1. **Choose a compact localization.** The local-ball existence lemma, applied to openness at x , permits choosing $R > 0$ with $\overline{B}(x, 4R) \subseteq U$. Put

$$B := B(x, R), \quad K := \overline{B}(x, 4R), \quad v(z) := \begin{cases} u(z), & z \in K, \\ 0, & z \notin K. \end{cases}$$

The inner ball is open, contains x , and lies in $K \subseteq U$. The closed ball K is compact and measurable. Local integrability on U gives integrability on K , so v is globally integrable and therefore locally integrable on E . In particular $v = u$ on B , also almost everywhere for $\mu|_B$.

2. **Use one shrinking sequence of kernels.** Choose normalized smooth bumps φ_k centered at zero with inner radius $a_k = R/(2(k+2))$ and outer radius $r_k = R/(k+2)$. In particular,

$$\varphi_k \geq 0, \quad \int_E \varphi_k d\mu = 1, \quad \overline{\{z : \varphi_k(z) \neq 0\}} \subseteq \overline{B}(0, r_k).$$

$$v_k(y) := \int_E \varphi_k(t) v(y-t) d\mu(t).$$

These functions are smooth by the compact-kernel convolution theorem. The radii satisfy $r_k \rightarrow 0$ and $r_k = 2a_k$. The normalized-bump convergence theorem for locally integrable functions therefore gives

$$v_k(y) \longrightarrow v(y) \quad \text{for } \mu\text{-almost every } y.$$

The hypotheses are precisely local integrability of v , shrinking outer radii, and a uniform bound 2 on the ratio of outer to inner radius.

3. **Make every mollification constant on the inner ball.** Apply the vanishing-directional-derivative theorem to this open U , this local ball data, the given local integrability, and the weak equation. It gives

$$Dv_k(y)e = 0 \quad (y \in B, e \in E).$$

Thus $Dv_k(y)$ is the zero linear functional. The ball B is convex, hence preconnected, and v_k is differentiable there. Thus the constancy theorem for each local mollification applies; its proof uses zero-derivative constancy on the open preconnected ball and supplies a real number c_k such that

$$v_k(y) = c_k \quad \text{for every } y \in B.$$

This is pointwise constancy of each smooth mollification, not yet a statement about the limit u .

4. **Choose a single point at which the whole sequence converges.** The nonempty open ball has positive Haar measure. Restrict the almost-everywhere convergence in Step 2 to B and intersect it with almost-everywhere membership in the measurable set B . The inner-ball convergence-point lemma therefore supplies one $y_0 \in B$ at which the entire sequence converges. Set $c = v(y_0)$. Then

$$c_k = v_k(y_0) \longrightarrow v(y_0) = c.$$

This fixed good point is the reason the constants converge. We neither assume that (c_k) converges nor choose a different exceptional-set witness for each k .

5. **Pass to almost every other point and undo localization.** For almost every y with respect to $\mu|_B$, Step 2 gives $v_k(y) \rightarrow v(y)$ and $y \in B$. Step 3 gives $v_k(y) = c_k = v_k(y_0)$ for every k . Step 4 and uniqueness of real limits therefore imply

$$v(y) = \lim_{k \rightarrow \infty} v_k(y) = \lim_{k \rightarrow \infty} c_k = c.$$

Since $v = u$ on B , this gives $u = c$ almost everywhere for $\mu|_B$. Taking $V = B$ supplies all the promised local data.

Main citations

- Exact declaration and its source — `MathlibAnnex.WeakDivergenceZero.nonempty_localAEConstantAt`
- Weak equation tested by divergence — `MathlibAnnex.WeakDivergenceZero`
- Vanishing derivatives for the localized mollification — `MathlibAnnex.WeakGradient.localMollification_fderiv_apply_eq_zero`
- Directional derivative formula — `MathlibAnnex.WeakGradient.localMollification_fderiv_apply`
- The six-field local almost-everywhere datum — `MathlibAnnex.LocalAEConstantAt`
- Local ball data: radius, positivity, and closed four-radius inclusion — `MathlibAnnex.WeakGradient.LocalBallData`
- The open inner ball — `MathlibAnnex.WeakGradient.LocalBallData.inner`
- The compact four-radius carrier — `MathlibAnnex.WeakGradient.LocalBallData.carrier`
- The closed middle ball — `MathlibAnnex.WeakGradient.LocalBallData.middle`
- The center belongs to the inner ball — `MathlibAnnex.WeakGradient.LocalBallData.center_mem_inner`
- Openness of the inner ball — `MathlibAnnex.WeakGradient.LocalBallData.isOpen_inner`
- Nonemptiness of the inner ball — `MathlibAnnex.WeakGradient.LocalBallData.inner_nonempty`
- Compactness of the localization carrier — `MathlibAnnex.WeakGradient.LocalBallData.isCompact_carrier`
- Measurability of the localization carrier — `MathlibAnnex.WeakGradient.LocalBallData.measurableSet_carrier`
- The carrier is contained in the domain — `MathlibAnnex.WeakGradient.LocalBallData.carrier_subset`
- The inner ball lies in the carrier — `MathlibAnnex.WeakGradient.LocalBallData.inner_subset_carrier`
- The middle ball lies in the carrier — `MathlibAnnex.WeakGradient.LocalBallData.middle_subset_carrier`
- The middle ball lies in the open domain — `MathlibAnnex.WeakGradient.LocalBallData.middle_subset_open`
- Choosing local ball data inside an open set — `MathlibAnnex.WeakGradient.nonempty_localBallData`
- Compact localization by an indicator — `MathlibAnnex.WeakGradient.compactLocalization`
- Localization agrees with the function on the carrier — `MathlibAnnex.WeakGradient.compactLocalization_of_mem`
- Localization is zero off the carrier — `MathlibAnnex.WeakGradient.compactLocalization_of_not_mem`
- Local integrability of the compact localization — `MathlibAnnex.WeakGradient.locallyIntegrable_compactLocalization`

- Almost-everywhere agreement on the inner ball —
MathlibAnnex.WeakGradient.compactLocalization_eq_ae_inner
- A uniform outer-to-inner radius ratio — MathlibAnnex.WeakGradient.shrinkingBump_ratio
- The outer radii tend to zero — MathlibAnnex.WeakGradient.tendsto_shrinkingBump_rOut_zero
- Almost-everywhere convergence of the local mollifications —
MathlibAnnex.WeakGradient.ae_tendsto_localMollification
- Zero in every direction gives the zero derivative —
MathlibAnnex.WeakGradient.localMollification_fderiv_eq_zero
- Smoothness of the localized mollification —
MathlibAnnex.WeakGradient.contDiff_localMollification
- Each mollification is constant on the inner ball —
MathlibAnnex.WeakGradient.exists_localMollification_constant
- Positive Haar measure of the inner ball —
MathlibAnnex.WeakGradient.localBall_inner_measure_pos
- A common convergence point in the inner ball —
MathlibAnnex.WeakGradient.exists_inner_convergencePoint
- Equality of every mollified value at two inner points —
MathlibAnnex.WeakGradient.localMollification_eq_at_inner_points
- Almost-everywhere constancy of the localized function —
MathlibAnnex.WeakGradient.compactLocalization_aeConstant_inner
- Transferring the constant back to the original function —
MathlibAnnex.WeakGradient.exists_aeConstant_inner
- Constructing all fields of the local constancy datum —
MathlibAnnex.WeakGradient.nonempty_localAECConstantAt

Lean source signature (exact)

```
theorem WeakDivergenceZero.nonempty_localAECConstantAt
  {U : Set E} {u : E → ℝ} (hweak : WeakDivergenceZero μ U u)
  (hU : IsOpen U) (hu : LocallyIntegrableOn u U μ)
  {x : E} (hx : x ∈ U) : Nonempty (LocalAECConstantAt μ U u x)
```

In the source	Mathematical meaning
$[NormedAddCommGroup\ E]\ [NormedSpace\ \mathbb{R}\ E]$ $[FiniteDimensional\ \mathbb{R}\ E]\ [MeasurableSpace\ E]\ [BorelSpace\ E]\ \{\mu : Measure\ E\}$ $[\mu.IsAddHaarMeasure]$	The finite-dimensional real normed space E , with Borel measurable structure and additive Haar measure μ . Dimension zero is allowed.
$(hweak : WeakDivergenceZero\ \mu\ U\ u)$	The scalar weak equation $\int_U u(z) \operatorname{div} W(z) d\mu(z) = 0$ for every C^1 vector field with a compact carrier contained in U ; $\operatorname{div} W = \operatorname{tr}(DW)$.
$(hU : IsOpen\ U)\ (hu : LocallyIntegrableOn\ u\ U\ \mu)$	The set U is open and u is locally integrable on U . Whole-space integrability or Lipschitz continuity of u is not assumed.

In the source	Mathematical meaning
$\{x : E\} \text{ (hx : } x \in U)$	The requested point $x \in U$, which itself may be exceptional for an a.e. equality.
<code>Nonempty (LocalAECostantAt μ U u x)</code>	The conclusion is existence of a local record, with the six fields decoded below; it does not choose an arbitrary previously specified neighbourhood.
<code>neighborhood</code>	The selected neighbourhood V of x ; this and the next five rows read the separately cited local record.
<code>isOpen_neighborhood</code>	V is open in the ambient topology.
<code>mem_neighborhood</code>	The specified point satisfies $x \in V$.
<code>neighborhood_subset</code>	$V \subseteq U$.

In the source	Mathematical meaning
<code>constant</code>	One value $c \in \mathbb{R}$ for this neighbourhood.
<code>ae_eq</code>	$u(y) = c$ for $(\mu _V)$ -almost every y . The same V, c occur in these six fields.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 7–12:

```
noncomputable section
open Set MeasureTheory TopologicalSpace
namespace MathlibAnnex
variable {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [FiniteDimensional ℝ E] [MeasurableSpace E] [BorelSpace E]
variable {μ : Measure E} [μ.IsAddHaarMeasure]
```

▼ Exact content identity

Declaration: `MathlibAnnex.WeakDivergenceZero.nonempty_localAECostantAt`

Accepted content SHA-256: a26a423bf64eda9884ab3d44deb8c12d8ab30eed12d3051d2626690322b0d158

Accepted source guide SHA-256: 87bd8f1e12997df114ea605364f4db3312b70b2166163623f6dc20359a69e14a

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73